Quantification: 701_z26c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	14
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1922.92	1500	1365.51	px^2
major_axis_length_px	lacuna shape	75.4752	67.0082	29.7312	px
minor_axis_length_px	lacuna shape	38.0255	32.9459	15.5667	px
aspect_ratio	lacuna shape	2.1407	1.9822	0.8206	unitless
eccentricity	lacuna shape	0.7964	0.8634	0.1957	unitless
solidity	lacuna shape	0.7414	0.7609	0.0869	unitless
roots_count	headline	10.1429	8	4.8494	count
ring_length_r30_px	headline	412.357	387	163.65	px
ring_length_r60_px	reported	975.286	999	400.357	px
owned_length_px	ownership-dependent	1546.62	979.015	1381.67	px
edge_count	ownership-dependent	105.786	65.5	94.922	count
mean_edge_length_px	ownership-dependent	14.7002	14.482	1.6868	px
perimeter_px	normalised	278.299	209.025	156.449	px
ring_area_r30_px2	normalised	8450.57	7637	2918.54	px^2
ring_area_r60_px2	normalised	20263.6	19812	5896.35	px^2
in_frame_fraction_r30	normalised	0.9961	1	0.0145	unitless
in_frame_fraction_r60	normalised	0.9868	1	0.0389	unitless
ring_density_r30	normalised	0.0489266	0.047078	0.01131	px^-1
ring_density_r60	normalised	0.0478159	0.0452979	0.0137652	px^-1
roots_per_100px_perimeter	normalised	3.863	3.8069	0.7371	per 100 px
ring_attached_length_r30_px	network v2	350.286	302	166.049	px
ring_attached_length_r60_px	network v2	674.5	734	418.183	px
ring_length_w_r30_px	network v2	466.118	432.948	184.414	px
ring_length_w_r60_px	network v2	1097.86	1129.41	449.628	px
sholl_crossings_r10	network v2	10.0714	8.5	4.6154	count
sholl_crossings_r20	network v2	10.8571	9.5	4.6386	count
sholl_crossings_r30	network v2	12.5	12	5.4737	count
loop_count	loops, report only	7.8571	1	10.9324	count
loop_edge_share	loops, report only	0.1671	0.062	0.1914	share
ring_width_mean_r30_px	width	6.2685	6.6518	1.1937	px

Per field
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measure	value	units
analysed_area_px2	762050	px^2
total_skeleton_length_px	37303	px
canalicular_length_density_per_px	0.0489509	px^-1
junction_count	1066	count
junction_density_per_px2	0.00139886	px^-2
lacuna_count	14	count

lacunae_per_px2	1.837e-05	px^-2
skeleton_component_count	668	count
mean_component_length_px	55.8428	px
median_component_length_px	13	px
field_length_density_w_per_px	0.0548718	px^-1
field_density_without_flagged_per_px	0.0336205	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	226	px
unattached_component_count	15	count
canalicular_length_density_attached_per_px	0.0489509	px^-1
field_length_density_with_unattached_per_px	0.0492474	px^-1
excluded_area_px2	261257	px^2
analysed_field_area_px2	787319	px^2
loop_count	186	count
loop_edge_share	0.2129	share
edge_count	3523	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.3932	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	29482	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2287	125.157	32.9459	3.7989	0.9647	0.6065	0.5498	75.4101	463.222	False
2	5110	133.797	67.4988	1.9822	0.8634	0.6757	1.2061	267.96	890.703	False
3	1782	62.6142	47.9495	1.3058	0.6431	0.682	1.1297	243.333	206.608	False
4	1422	68.8124	28.5123	2.4134	0.9101	0.8335	1.1259	314.225	198.443	False
5	1652	67.0082	31.8663	2.1028	0.8797	0.8959	0.9049	359.245	256.938	False
6	967	67.0717	21.9089	3.0614	0.9451	0.7662	-1.3191	319.159	644.63	False
7	821	63.8935	20.0569	3.1856	0.9495	0.7609	0.1039	594.211	741.577	False
8	2272	79.8087	42.6154	1.8728	0.8455	0.7881	-1.0855	686.797	269.451	False
9	656	33.1626	30.2759	1.0953	0.4081	0.801	1.2644	753.977	863.463	False
10	1435	61.7195	33.8344	1.8242	0.8363	0.8112	-1.4959	787.26	645.571	False
11	4376	111.203	63.8724	1.741	0.8186	0.6554	-1.0966	845.546	482.61	False
12	1500	55.4417	51.2017	1.0828	0.3835	0.6268	-1.2338	890.901	936.497	False
13	718	51.4892	21.7937	2.3626	0.906	0.7349	-1.184	917.613	557.102	False
14								904.956	141.983	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	19	555	1240	748.027	57	13.1233	6.735
2	18	560	1371	2113.85	142	14.8863	6.7296
3	12	400	707	298.108	22	13.5503	5.6004

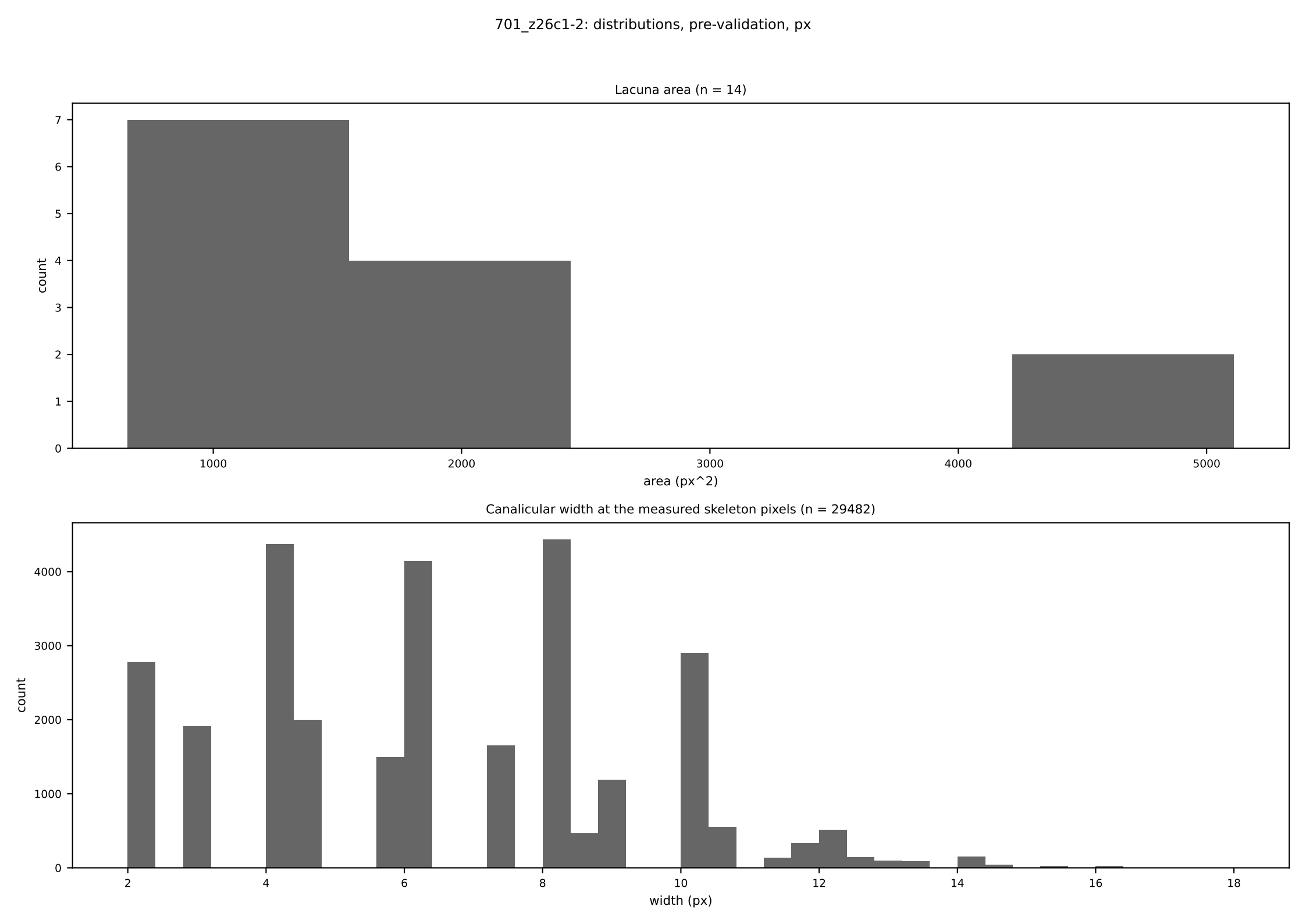
4	7	213	361	238.794	17	14.0467	4.2883
5	6	233	671	258.421	13	19.8786	3.9191
6	8	304	690	475.576	34	13.9875	6.574
7	7	334	887	856.039	56	15.2864	7.0448
8	8	374	1082	2572.48	169	15.2218	6.8797
9	7	259	461	303.208	24	12.6337	5.4386
10	8	484	1125	2006.26	140	14.3304	7.4781
11	18	808	1917	4506.14	316	14.26	5.869
12	12	475	1074	1101.99	74	14.8918	5.9812
13	7	497	1144	2456.85	163	15.0727	8.3653
14	5	277	924	3716.92	254	14.6336	6.8562

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	459.683	11984	25966	0.9457	0.8575	0.0463117	0.0477548	4.1333
2	541.524	14453	33199	1	1	0.0387463	0.0412964	3.324
3	345.031	7948	17293	1	1	0.0503271	0.0408836	3.4779
4	202.35	5401	10020	1	1	0.0394371	0.0360279	3.4593
5	196.522	6388	15414	1	1	0.0364746	0.0435318	3.0531
6	188.244	7408	20330	1	1	0.0410367	0.03394	4.2498
7	163.723	6981	19532	1	1	0.0478442	0.0454127	4.2755
8	273.019	9285	23947	1	1	0.04028	0.0451831	2.9302
9	133.338	6168	17509	1	1	0.0419909	0.0263293	5.2498
10	215.7	7866	20092	1	1	0.0615306	0.0559924	3.7089
11	605.222	13350	26836	1	1	0.0605243	0.0714339	2.9741
12	307.298	9361	22441	1	0.9574	0.0507424	0.0478588	3.905
13	167.823	6421	14897	1	1	0.0774023	0.076794	4.1711
14	96.7107	5294	16215	1	1	0.0523234	0.0569843	5.1701

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	494	742	631.843	1398.45	20	10	12
2	520	944	638.299	1549.24	16	18	17
3	238	266	441.706	783.612	9	8	10
4	162	206	241.622	413.031	7	4	6
5	146	182	265.664	769.696	6	7	7
6	215	344	336.936	767.512	6	7	9
7	271	461	388.433	1002.41	7	9	15
8	333	760	424.191	1209.62	8	10	13
9	220	284	295.522	508.889	5	8	4
10	448	1002	540.99	1257.48	11	15	18
11	716	1663	908.654	2145.81	17	20	24
12	407	726	538.103	1213.09	11	13	18
13	490	1061	563.132	1302.05	11	15	12
14	244	802	310.551	1049.19	7	8	10


701_z26c1-2: distributions, pre-validation, px
Lacuna area (n = 14)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 29482)
count
width (px)